REPORT - ERP PROJECT

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1. Introduction

This project implements a University ERP System with three distinct user roles - Admin, Instructor and Student. There are separate authentication and ERP databases. The application enables login based access, enrollment management, grade entry, maintenance mode control and secure operations like create, read, delete for academic data. We have used the Java Swing interface backed by MySQL.

2. System Overview

The ERP system is divided into two major components :

(i) Authentication Database (university auth)

This stores user login details, roles and hashed passwords. It also links users to their identities in the ERP database.

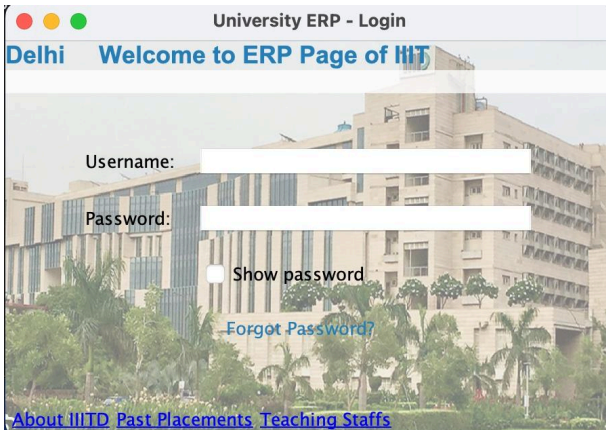
(ii) ERP Database (university_erp)

This stores all academic domain data such as students, instructors, courses, sections, enrollments, grades and system settings. It also contains the maintenance mode flag which controls access across the application.

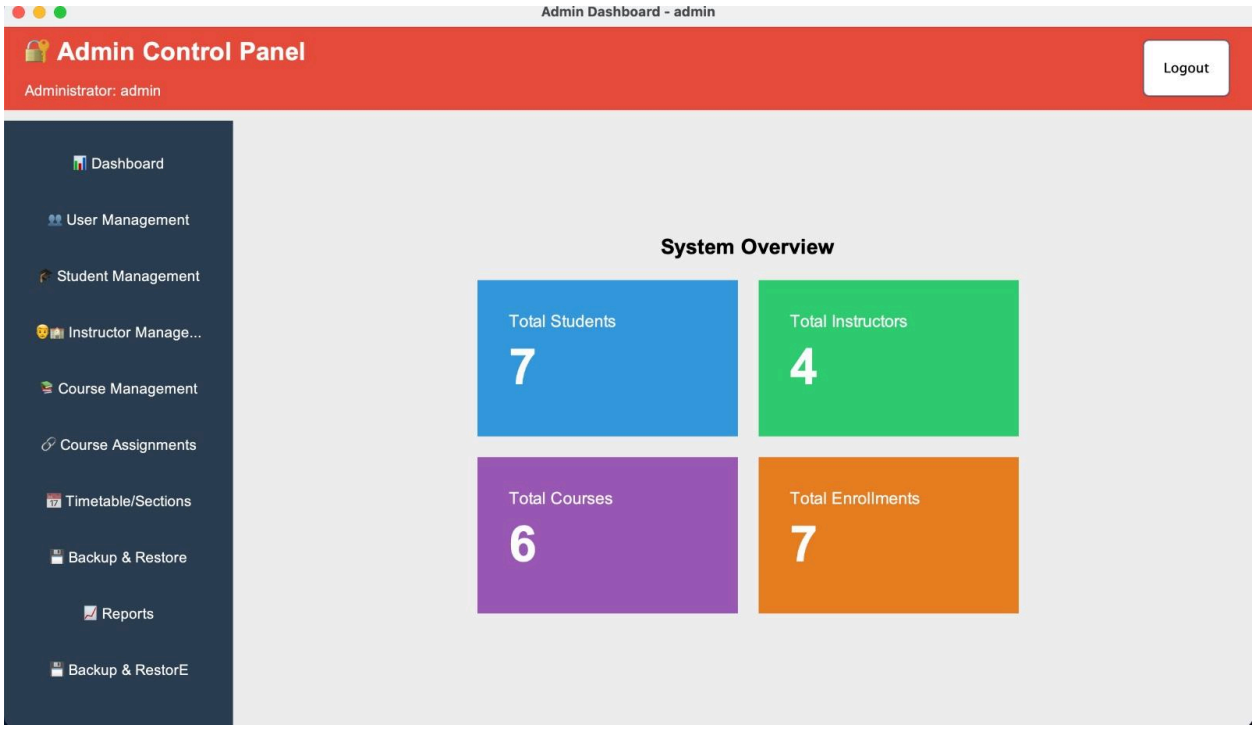
The UI is built using Java Swing. MySQL is accessed via JDBC. Login validation uses BCrypt password hashing for secure authentication.

3. Application Screenshots

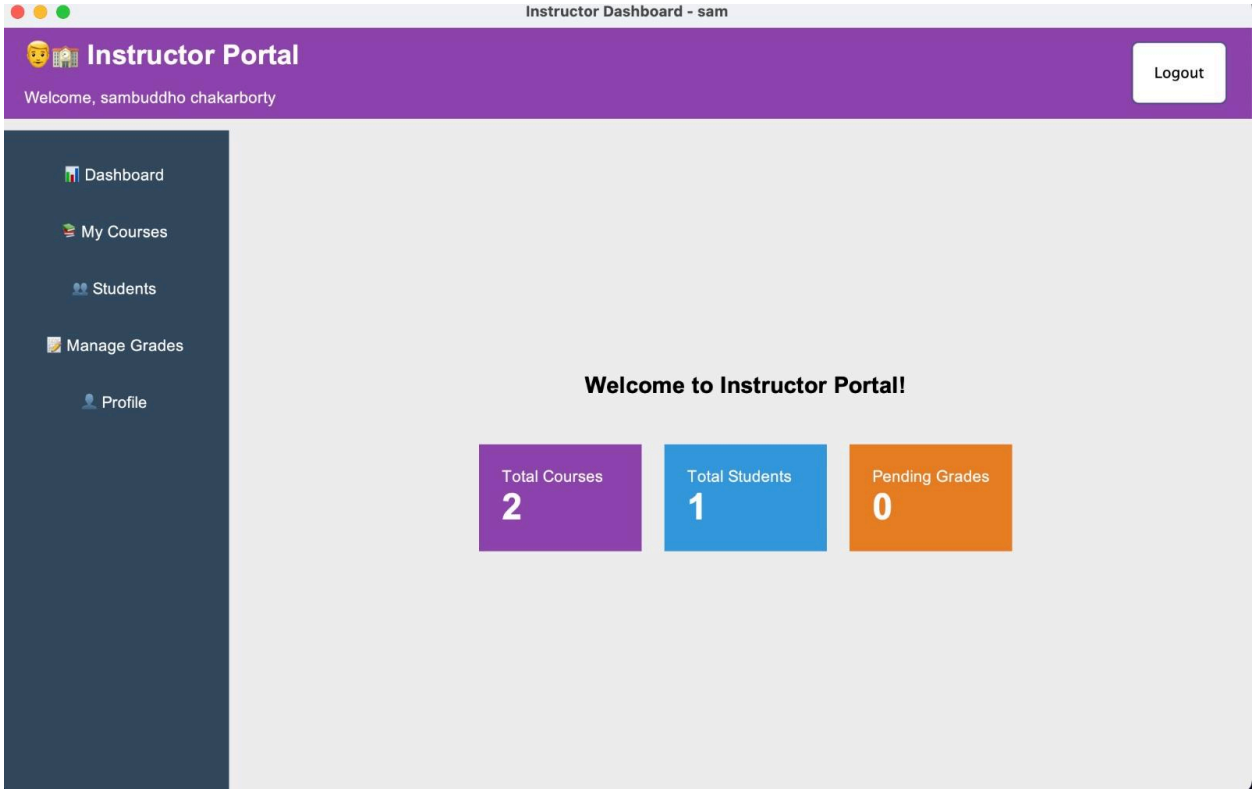
(i) Login Screen



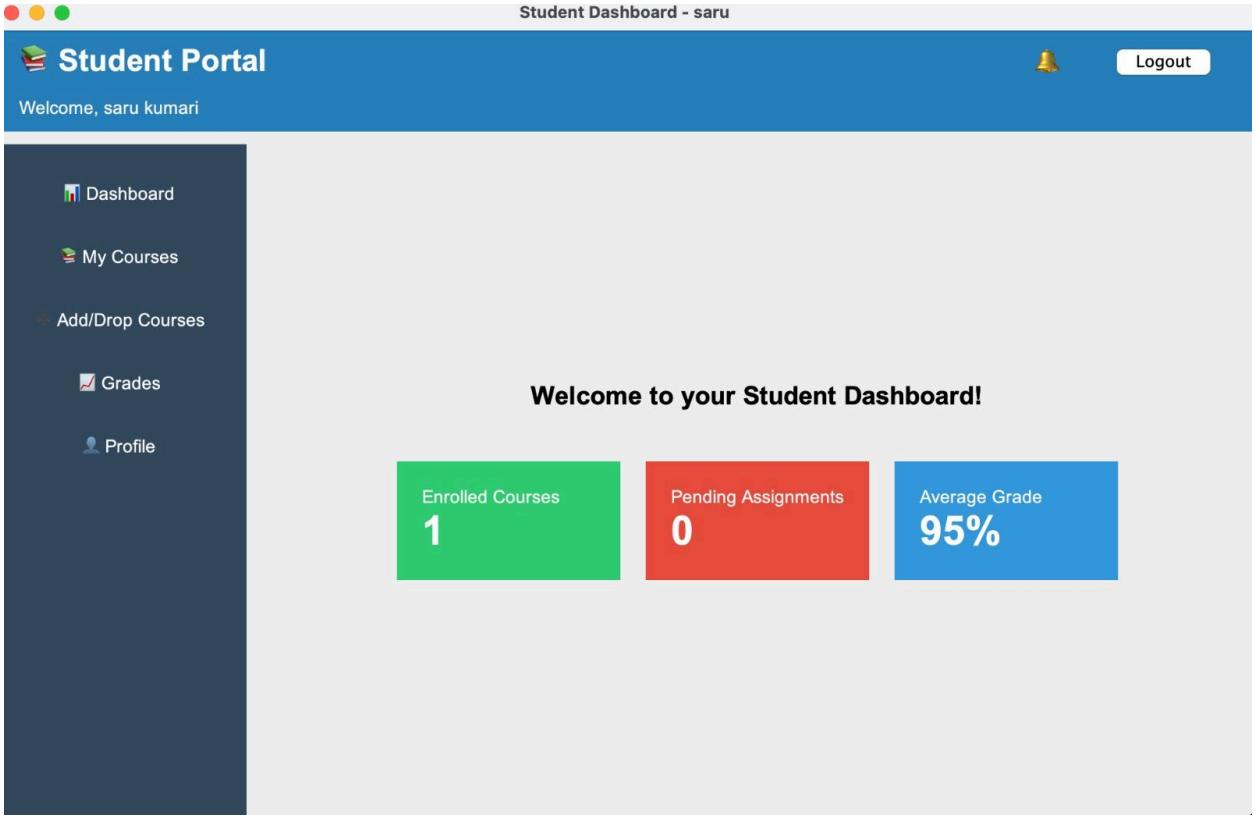
(ii) Admin Dashboard



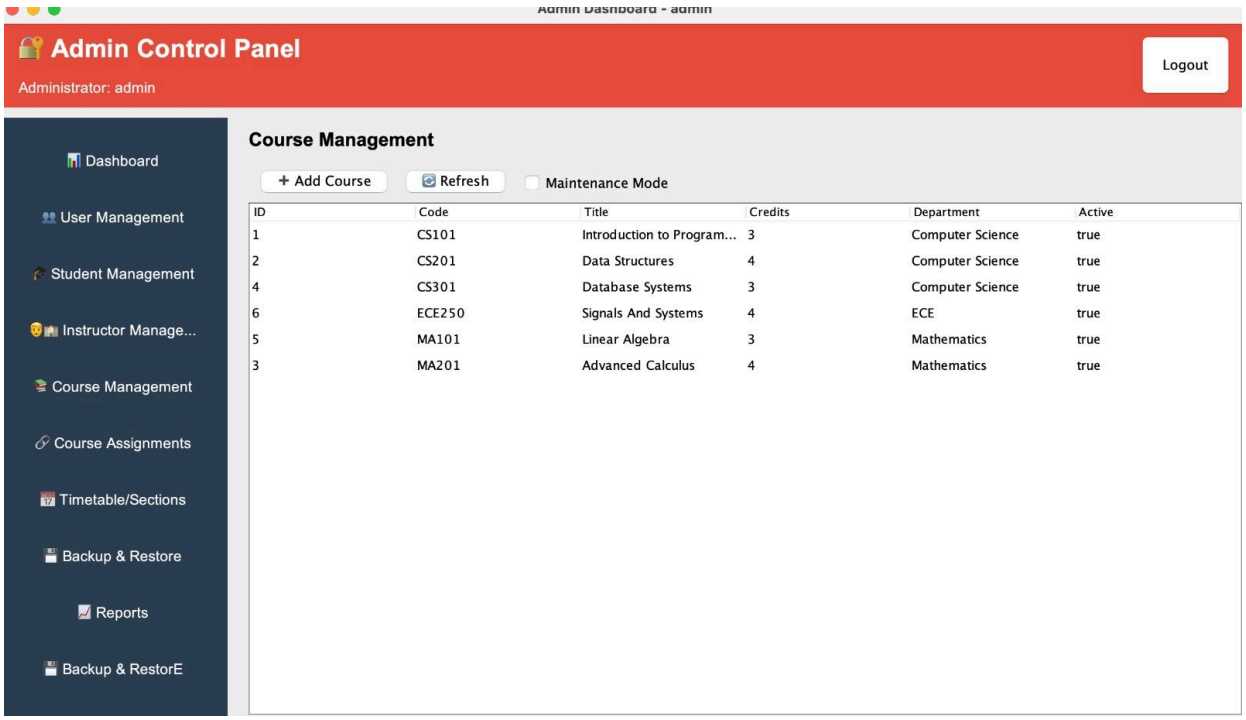
(iii) Instructor Dashboard



(iv) Student Dashboard



(v) Course List



4. Final Grade Weighting Rule

The system calculates final student grades based on weighted assessments.

Each grade record stores:

- assessment_type
- obtained_score
- max_score
- weight (percentage)

Example:
If a student has:

- Quiz: 18/20 weighted 20%
- Midterm: 40/50 weighted 30%
- Final Exam: 75/100 weighted 50%

Final Grade =

(0.90×20)+(0.80×30)+(0.75×50)=82.5(0.90 \times 20) + (0.80 \times 30) + (0.75 \times 50) = 82.5(0.90×20)+(0.80×30)+(0.75×50)=82.5

The instructor may enter grades manually or upload a CSV file, after which the system automatically applies the weighting rule and stores values in the grades_batch table.

5. Enforcement of Roles & Maintenance

The users_auth table assigns each user a role:

- Admin : Manages users, modify courses, create sections, toggle maintenance mode, view all data
- Instructor : View assigned sections, upload/ enter grades, view enrolled students
- Student : View courses, view grades, request or see enrollment

The settings_batch table contains:

- maintenance_mode (true/false)
- current_semester
- current_year

When maintenance mode is on:

- Students cannot enroll in or drop sections.

- Instructors cannot edit grades or upload CSV.
- Admin retains full access.

6. Database Table Lists

(i) Authentication Database : university_auth

This stores user ID, username, hashed password and role (users_auth),. It also tracks failed login attempts.

(ii) ERP Database : university_erp

Table	Purpose
students_bach	Student records + link to auth user
instructors_bach	Instructor records + link to auth user
courses_bach	Courses Offered
sections_bach	Course sections assigned to instructors
enrollments_bach	Student-section enrollments
grades_bach	Individual graded assessments
notifications_bach	Alerts for users
settings_bach	System settings including maintenance mode

7. EXTRAS ADDED

Additional functionalities incorporated into the system include a password update module, an internal notification mechanism, transcript generation/export, full ERP database backup and restore, and an account lockout triggered after three consecutive failed login attempts.